Dmitrii Vasilenko

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Research Interests

Theoretical Machine Learning, Bayesian Deep Learning, Neural Network Dynamics.

Education

Moscow Institute of Physics and Technology (MIPT) — Moscow, Russia
*M.Sc. in Computer Science (GPA: **4.9/5.0**)* — *Sep. 2025 – Present*

Yandex School of Data Analysis (YSDA) — Moscow, Russia
*M.Sc. equivalent in Data Science (GPA: **5.0/5.0**)* — *Sep. 2025 – Present*

Moscow Institute of Physics and Technology (MIPT) — Moscow, Russia
*B.Sc. in Applied Mathematics and Physics (GPA: **4.5/5.0**)* — *Sep. 2021 – July 2025*

Publications

Jacobian Analysis of a Recurrent Transformer Block | *AINL 2026* — Apr. 2026

- D. Vasilenko, I. Stepanov, V. Kasiuk, G. Karpeev, A. Grabovoy.
- Accepted to Artificial Intelligence and Natural Language Conference (AINL 2026).
- Derived explicit upper bounds on the spectral norm of the full-step Jacobian for Recurrent Transformers.
- Theoretically proved and empirically validated that RMSNorm acts as a multiplicative damping factor, preventing gradient explosion and maintaining an $\mathcal{O}(1)$ Jacobian norm.

Bensemble: A Modular Python Library for Bayesian Deep Learning | *JOSS (submitted)* | GitHub — Jul. 2026

- D. Vasilenko, M. Nabiev, F. Sobolevsky, V. Kasyuk, O. Bakhteev.
- Open-source PyTorch library providing a unified interface for 8 Bayesian deep learning and neural network ensembling methods.
- Implements Deep Ensembles, MC Dropout, Variational Inference, Laplace Approximation, Probabilistic Backpropagation, and Neural Ensemble Search.
- Validated cross-method correctness on CIFAR-10 vs. SVHN out-of-distribution detection benchmarks.

Research Experience

Yandex R&D, YandexGPT Pretraining Team — Moscow, Russia
Research Intern — *Aug. 2026 – Present*

- Conducting research on LLM pretraining within the Knowledge Research group.
- Investigating reasoning capabilities and knowledge acquisition in large language models.

Intelligent Systems Department, MIPT — Moscow, Russia
Graduate Researcher — *Sep. 2025 – Present*

- Researching theoretical stability and dynamics of Recurrent Transformers under the supervision of Dr. Andrey Grabovoy.
- Investigating Jacobian dynamics and spectral properties of deep Transformer components (MSA, FFN, RMSNorm).

Moscow Institute of Physics and Technology (MIPT) — Moscow, Russia
Undergraduate Researcher — *Apr. 2024 – July 2025*

- Developed and implemented numerical solvers in C++ for hyperbolic partial differential equations as part of the Bachelor's thesis.
- Analyzed systems of nonlinear equations using the Method of Characteristics, demonstrating strong proficiency in classical numerical methods and computational mathematics.

Technical Skills

Programming: Python, C++

ML Frameworks: PyTorch, NumPy

Developer Tools: Git, Docker, Linux, uv, LaTeX, GitHub Actions, pytest, Ruff

Natural Languages: Russian (Native), English (Advanced / C1), German (A2)